

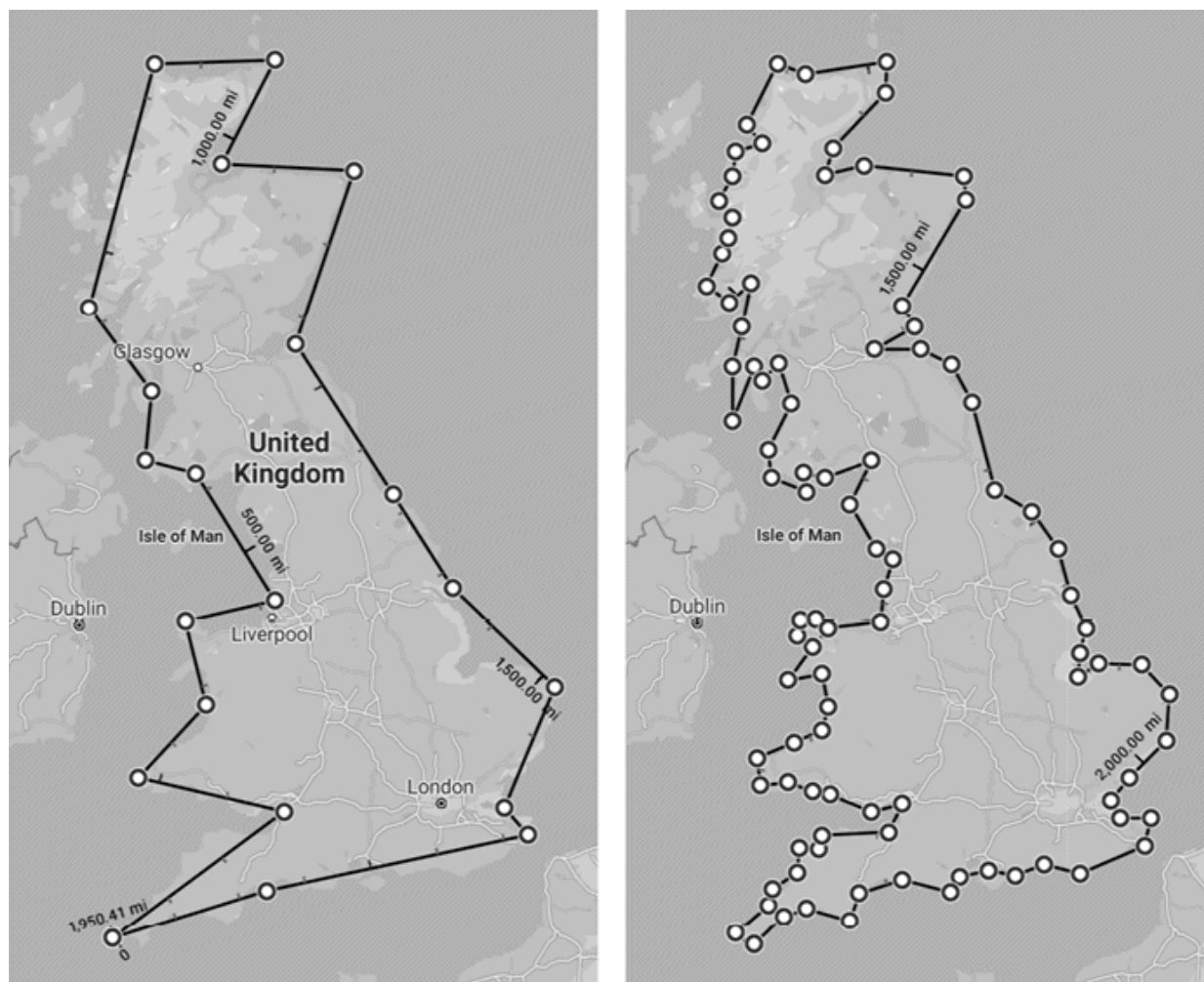


Figure 9-13: The island of Great Britain, with a rough measure (left) and more precise measure (right). Measuring the coast more precisely adds 800 miles to its length.

Mandelbrot's key insight about fractals such as the coastline of Britain is that you can continue to look closer and closer, and there will continue to be "roughness" at every scale. So, as your measurement gets finer and finer, the length of the coastline will get longer and longer. The "coast" will follow the Thames upriver, deep into the landmass along one bank and back out to the English Channel on the other bank. Thus, the answer to our question of Great Britain's coastline's length is, "It depends."

The *Koch curve* fractal has a similar property pertaining to the length of its coastline, or rather, perimeter. First introduced in 1902 by Swedish mathematician Helge von Koch, the Koch curve is one of the earliest fractals to be described mathematically. To construct it, take a line of length b and divide it into three equal parts, each of length $b / 3$. Replace the middle